

A2 · It Is Always DNS (Except When It Is Not)

CS425 Computer Networks · in class activity · 45 minutes · 20 points

TURN THIS IN ON PAPER
before you leave the room
nothing is accepted afterward

Names _____ Date _____

TARGET CARD · copy these from the board before you touch anything

ALPHA = _____ BRAVO = _____

YOUR FOUR PROBES

QUESTION	MACOS AND LINUX	WINDOWS
Does the name resolve?	<code>dig +short NAME @ALPHA</code>	<code>nslookup NAME ALPHA</code>
Does the host answer ICMP?	<code>ping -c 3 ADDR</code>	<code>ping -n 3 ADDR</code>
Is the port open?	<code>nc -vz -w 5 ADDR PORT</code>	<code>Test-NetConnection ADDR -Port PORT</code>
Does the service work?	<code>curl -sS -o /dev/null -m 8 -w 'connect=%{time_connect} total=%{time_total} code=%{http_code}\n' http://ADDR:PORT/</code>	

Read the words **curl prints when it fails**. There is a large difference between **Connection refused**, **Connection timed out**, and **Could not resolve host**. Telling those three apart is the whole activity. If your **dig** answers disagree with your neighbor's, your network is intercepting port 53: add `-p 5353` and write that down, it is a real finding.

ROUND 1 · THE HEALTHY SIGNATURE

5 MIN · 2 PTS

You cannot recognize broken until you have looked at working. Probe ALPHA port 8080 with all four commands.

#	TARGET	DIG SAID	PING SAID	CURL PRINTED, WORD FOR WORD	SECONDS TO ANSWER
1	ALPHA 8080 baseline				

Answer before you move on

What did ping report as the minimum round trip time, what did curl report for **connect=**, and why are those two numbers in the same neighborhood?

ROUND 2 · FOUR WAYS TO FAIL

20 MIN · 11 PTS · FIVE MINUTES PER STATION

Do them in order. The order is the argument. Stations 2 and 3 are the same host, the same laptop and the same network, and they behave in opposite ways.

Fill in the table overleaf. For each station write what each probe actually printed, how many seconds it took to give up, the cause in your own words, and which layer the fault is at.

ROUND 2 · STATION TABLE

#	TARGET	DIG SAID	PING SAID	CURL PRINTED, WORD FOR WORD	SEC TO FAIL	THE CAUSE, IN YOUR WORDS	LAYER
2	ALPHA 8081						
3	ALPHA 8082						
4	ghost .cs425.lab						
5	mirage .cs425.lab						
6	BRAVO 8080						
7	ALPHA 8084 stretch						

The one distinction that matters

5 pts

Stations 2 and 3 are the two failures you will meet for the rest of your career, and they are opposites. One means a machine received your packet and answered “no”. The other means nobody answered anything at all.

Which is which, and **what did the host send back** in the case that failed fast? It is one TCP flag.

Station 4

3 pts

What was in the **status:** field of the answer, and is it even *possible* for this to be a connectivity problem? How many packets did you send to the target?

Station 5

3 pts

This looks identical to station 2 from where you are sitting, and it is not the same failure at all. Name the **one probe** that separates them, and say what is special about the address dig handed you.

ROUND 3 · THE PROBE THAT LIES

8 MIN · 3 PTS

Ping BRAVO. Then curl BRAVO port 8080. One fails, one succeeds, and the one that fails is the one most people would have used to decide the host was dead.

1. Which probe reported the machine unreachable, and which one proved it was not?

2. What protocol does ping use, and at which layer does that protocol live?

3. ALPHA answers ping and BRAVO does not, yet both serve the same page on 8080. What is different between the two hosts, and does that difference live *on* the host or *in front of* it?

4. Write one sentence you could say to a coworker who has just told you “the server is down, it does not ping”.

ROUND 4 · BUILD THE DECISION TREE

7 MIN · 3 PTS

This is the part you keep. Every leaf must name a layer and name who can fix it: you, the person running the server, or the network in between.

START · “it does not work”. First question, because if the answer is bad you never sent a single packet to the target:

No answer / NXDOMAIN.

cause _____ layer _____ fixed by _____

An answer comes back, but it is an address you cannot route to.

cause _____ layer _____ fixed by _____

An answer comes back and it is routable. Next question: _____

Fails instantly. The word curl uses is _____. The host sent back a _____.

cause _____ layer _____ fixed by _____

Hangs, then gives up after about _____ seconds. What came back? _____

cause _____ layer _____ fixed by _____

The connection completes. Last question: _____

Open, but no bytes ever arrive.

cause _____ layer _____ fixed by _____

It works. If ping *still* fails, that tells you _____

ROUND 4 · SUMMARY TABLE

One line per symptom. This is the sheet you photograph and keep.

STA	WHAT YOU SAW	WHAT IT MEANS ABOUT WHAT THE OTHER END DID	LAYER	WHO FIXES IT
1				
2				
3				
4				
5				
6				
7				

ROUND 5 · SAY IT OUT LOUD

5 MIN · 1 PT

1. Give the two symptom words that a hang and a refusal produce, and say what the server sends in each case. One of them sends nothing at all.

2. **You are on call.** Somebody reports “our API is down”. You get *one* command before you have to say something useful. Which single command do you run, and what are the three different things its output can tell you? Defend the choice.

IF YOU FINISH EARLY

Station 7, ALPHA 8084. The connection succeeds and then nothing ever comes back. `nc -vz` is delighted; curl sits there until it gives up. Which layer completed successfully and which one never did? Why is a health check that only tests whether the port opens worse than useless here?

The other side. For station 2 your packets vanished. Did they vanish before they reached the host or after? You cannot tell from your laptop, which is the honest and uncomfortable answer. Write the command you would run *on the server* to find out whether your SYN ever arrived.

GRADING · 20 POINTS TOTAL	
Baseline row has real numbers, and connect= is related to the ping RTT	2
Stations 2 and 3 distinguished correctly, and the TCP flag behind the fast failure is named	5
Station 4 identified as name resolution, not connectivity, with the status field as evidence	3
Station 5 separated from station 2, with a reason the returned address cannot be routed	3
Round 3 correct, including the layer ICMP lives at and why a failed ping is not a dead host	3
Decision tree complete, every leaf names a layer and an owner, summary table matches	3
Round 5 question 2 picks a defensible command and gives three distinct outcomes	1

Hand this in before you leave. Every name at the top.